Such philosophical and religious musings on space can be compelling and provocative, yet, as in Einstein's cautionary remark above, they lack a critical sharpness of description. But there *is* a fundamental and precisely framed question that emerges from such discourse: should we ascribe an independent reality to space, as we do for other, more ordinary material objects like the book you are now holding, or should we think of space as merely a language for describing relationships between ordinary material objects?

The great German philosopher Gottfried Wilhelm von Leibniz, who was Newton's contemporary, firmly believed that space does not exist in any conventional sense. Talk of space, he claimed, is nothing more than an easy and convenient way of encoding where things are relative to one another. Without the objects *in* space, Leibniz declared, space itself has no independent meaning or existence. Think of the English alphabet. It provides an order for twenty-six letters—it provides relations such as *a* is next to *b*, *d* is six letters before *j*, *x* is three letters after *u*, and so on. But without the letters, the alphabet has no meaning—it has no "supra-letter," independent existence. Instead, the alphabet comes into being with the letters whose lexicographic relations it supplies. Leibniz claimed that the same is true for space: Space has no meaning beyond providing the natural language for discussing the relationship between one object's location and another. According to Leibniz, if all objects were removed from space—if space were completely empty—it would be as meaningless as an alphabet that's missing its letters.

Leibniz put forward a number of arguments in support of this so-called *relationist* position. For example, he argued that if space really exists as an entity, as a background substance, God would have had to choose where in this substance to place the universe. But how could God, whose decisions all have sound justification and are never random or haphazard, have possibly distinguished one location in the uniform void of empty space from another, as they are all alike? To the scientifically receptive ear, this argument sounds tinny. However, if we remove the theological element, as Leibniz himself did in other arguments he put forward, we are left with thorny issues: What is the location of the universe within space? If the universe were to move as a whole—leaving all relative positions of material objects intact—ten feet to the left or right, how would we know? What is the speed of the entire